THE FIDGET CRIMPER

An ADHD Augmentation Device

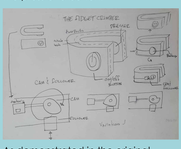
As a person with ADHD, fidget toys are a big part of my life. I have used every kind, from clickers to flippers to spinners. The problem with these devices is that they can become their own distraction if they are too fun, and they require a user to physically use them.

The Fidget Crimper solves both of these problems. This device is designed to augment a user's ability to focus in any situation. The Fidget Crimper fits on a user's hand and applies constant pressure. This reminds the user that there is a task at hand so they can redirect their focus.

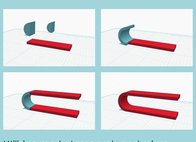
This device is designed specifically to improve attention span for the user. Its ability to augment a user's focus is based on how ADHD actually works. Normal fidget toys function as an outlet for an overactive nervous system, while the Fidget Crimper actually redirects the user's focus back to the task at hand.

The Process

The first step in developing the fidget crimper began with conception. I discovered that having a person squeeze my hand at times I need to focus was highly effective, so I invented a device that could provide the same effect. The Fidget Crimper is a small, plastic device that fits on to a user's hand. The original design used a simple mechanical device to squeeze the user's hand in intervals, toggled by the press of a button.



As demonstrated in the original blueprints, the original design was complex. I wanted to create an electronic device that used a cam and follower to continuously press into the user's hand, toggled by a button on the outside. Due to a lack of knowledge in mechanical engineering, I decided to change my approach. I began creating the new design in TinkerCad, a free 3D modeling tool on the internet. This design was a simple machine that could be created with a 3D printer. From the pictures below, you can see the creation process from beginning to end. The first version had hard edges, and only one size.



Utilizing market research and micro usability experiments, I discovered that my original design had several flaws. The biggest issue was that the hard rectangular edges caused several problems. For larger hand sizes, the edges caused discomfort. Additionally, they made the Fidget Crimper difficult to wear in certain positions for prolonged periods of time.

The second problem was that there was only one size. I realized that people of all ages could benefit from this device, so I wanted to solve that problem as well. These revelations led me to design the second iteration, shown below. This version features a rounded edge on the top and bottom, as well as several sizes. The Fidget Crimper attached to the display is the large size, indicated by the "L" on the side. I also designed a logo and chose a font to improve the branding of the product.

Accessibility, Usability, & User Experience

The real beauty of the Fidget Crimper lies in its simplicity. In order to truly augment a user's experience with the symptoms of ADHD, something more than an outlet is required. To better understand the product, it is important to understand ADHD. ADHD is a neurodevelopmental condition in which the brain's reward and attention systems are wired differently, making it difficult to regulate focus, control impulses, and manage physical energy. Fidget toys help the brain concentrate and regulate emotions by giving the nervous system controlled, repetitive sensory input. By keeping the hands occupied, they absorb excess physical energy or soothe anxiety, freeing up one's cognitive resources so the individual can focus on primary tasks.

Although this can be an effective solution, there are several problems with typical fidget toy designs. One of the biggest is that they require a user's hands to function. If a user with ADHD is working on anything that requires the use of both hands, they are working in intermittent periods of focus. There are foot-only fidget options or active sitting tools, but these are large and can only be used in a sitting position.

The second big issue with fidget toys is that they give an impression. Adults in the professional world might not want to rely on something that looks like a toy to function at the same level as their peers. The Fidget Crimper is a simple, elegant design that can be worn with any outfit, from street clothes to suits. Like a wrist watch, it is functional and fashionable.

In Brandon Strubberg's "Diversity Topics: AI, Accessibility, and Augmentation" class last semester, I gained knowledge of the emerging technologies of today. Through the examination of how technology affects our lives at every level, I began to see the potential of 3D printers. Artists all over the country are starting small businesses with their printers, using them to design and fabricate products that augment a user's experience with many aspects of life. Seeing this, I realized that I could do the same.

By employing a digital-humanities method of critical making and reflective analysis to explore embodiment, accessibility, and human–technology relationships, the future of the Fidget Crimper became clear to me. 3D printing is an amazing technology that opens up all sorts of opportunities. As the proud owner of a new Bambu 2XD, I plan on using a combination of user experience testing, audience analysis, and further research into augmentation and emerging technologies to develop the Fidget Crimper into a product I can eventually take to market.

Thanks!